Type MS · Finding unknown constants from binomial coefficients

Answer 1 of 13 · Question 5 part b · 6 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

number		
5 (a)	$1+anx+\frac{n(n-1)}{2}a^2x^2+\frac{n(n-1)(n-2)}{3!}a^3x^3$	M1 A1 (2)
(b)	$an=15$	B1
	$\frac{n(n-1)}{2}a^2=\frac{n(n-1)(n-2)}{6}a^3\Rightarrow\left(3=(n-2)a\right)$	M1A1
	Solving simultaneously leading to $a=\ldots$ or $n=\ldots$	M1
	$a=6$ and $n=2.5$	A1 A1 (6)
(c)	$\frac{'2.5'\times1.5\times0.5\times'6'^3}{6}=67.5$	M1 A1 (2)
		Total 10 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 1 of 13 · Question 5 part b · 6 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 2 of 13 · Question 2 part a · 6 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

	$C\sqrt{2}+D$ e.g., $\frac{5+\sqrt{18}}{\sqrt{50}-6}\times\frac{\sqrt{50}+6}{\sqrt{50}+6}$ scores this mark. That is all that is required for this mark. Ignore $>$, $<$, $=$ for this mark
dM1	For explicitly multiplying out and simplifying their expression to obtain a value for x in the required form. Allow no more than one error in this simplification. However, it must be simplified as far as $\frac{A+\sqrt{B}}{C}$ or $\frac{P+Q\sqrt{R}}{T}$ where A, B, C, P, Q, R and T are integers Ignore $>$, $<$, $=$ for this mark
A1	For obtaining the value of x as given with the correct inequality. For candidates who give an estimated value [8.629….] , please isw if you see the exact value first.

Type MS · Finding unknown constants from binomial coefficients

Answer 2 of 13 · Question 2 part a · 6 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

ALT 1 – Squares both sides [Ignore <, >, = for the first 3 marks]	
M1	Squares both sides without errors. $\left(\sqrt{50}x-\sqrt{18}\right)^2>\left(6x+5\right)^2\Rightarrow 50x^2-60x+18>36x^2+60x+25$
M1	Collects up their like terms and forms a 3TQ $\Rightarrow 14x^2-120x-7>0$
dM1	Solves their 3TQ by either valid method using the formula or completing the square. $x=\frac{60\pm 43\sqrt{2}}{14}$ oe eg., $x=\frac{60\pm\sqrt{3698}}{14}$ Accept $\pm$ for this mark We MUST see a method for the award of this mark. Do not award for roots appearing with no working. Ignore >, <, = for this mark
A1	For obtaining the value of x as given with the correct inequality. This has to be $x>\frac{43\sqrt{2}+60}{14}$ For candidates who give an estimated value [8.629….] , please isw if you see the exact value first.
ALT 2 – Collects up like terms and squares [Ignore <, >, = for the first 3 marks]	
M1	Squares both sides without error. $\left(\sqrt{50}x-6x\right)^2>\left(5+\sqrt{18}\right)^2\Rightarrow 50x^2-12\sqrt{50}x^2+36x^2>25+10\sqrt{18}+18$
M1	Collects up like terms and forms a **2TQ** $86x^2-12\sqrt{50}x^2-\left(43+10\sqrt{18}\right)>0\Rightarrow\left[\left(86-60\sqrt{2}\right)x^2-\left(43+30\sqrt{18}\right)>0\right]$
dM1	Solves their 2TQ by either valid method using the formula or completing the square. $x=\frac{43\sqrt{2}\pm 60}{14}$ oe eg., $x=\frac{\sqrt{3698}\pm 60}{14}$ We MUST see a method for the award of this mark. Do not award for roots appearing with no working.
A1	For obtaining the value of x as given with the correct inequality. This has to be $x>\frac{43\sqrt{2}+60}{14}$ For candidates who give an estimated value [8.629….] , please isw if you see the exact value first.

Type MS · Finding unknown constants from binomial coefficients

Answer 2 of 13 · Question 2 part a · 6 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks
2(a)	$(1+Ax)^n = 1+n(Ax)+\dfrac{n(n-1)(Ax)^2}{2}+\ldots$	B1
	$\Rightarrow nA=-\dfrac{1}{3}$ $\quad \dfrac{n(n-1)A^2}{2}=\dfrac{5}{36}$	M1
	$\Rightarrow A=-\dfrac{1}{3n}$	
	$\Rightarrow \dfrac{n(n-1)\left(-\dfrac{1}{3n}\right)^2}{2}=\dfrac{5}{36}\Rightarrow n=-\dfrac{2}{3}\Rightarrow A=\dfrac{1}{2}$	M1dM1A1A1
	ALT 1	[6]
	$\Rightarrow nA=-\dfrac{1}{3}$ $\quad \dfrac{n(n-1)A^2}{2}=\dfrac{5}{36}$	[M1
	$\Rightarrow n=-\dfrac{1}{3A}$	
	$\Rightarrow \dfrac{-\dfrac{1}{3A}\left(-\dfrac{1}{3A}-1\right)(A)^2}{2}=\dfrac{5}{36}\Rightarrow A=\dfrac{1}{2}\Rightarrow n=-\dfrac{2}{3}$	M1dM1A1A1]
	ALT 2	
	$\Rightarrow nA=-\dfrac{1}{3}\Rightarrow\left[(nA)^2=\dfrac{1}{9}\right]$	
	$\dfrac{n(n-1)A^2}{2!}=\dfrac{5}{36}\Rightarrow\left[\dfrac{(nA)^2-A(An)}{2!}=\dfrac{5}{36}\right]$	[M1
	$\Rightarrow \dfrac{\dfrac{1}{9}-A\left(-\dfrac{1}{3}\right)}{2}=\dfrac{5}{36}\Rightarrow A=\dfrac{1}{2}\Rightarrow n=-\dfrac{2}{3}$	M1dM1A1A1]
(b)	Coefficient of $x^3 \Rightarrow \dfrac{\left(-\dfrac{2}{3}\right)\left(-\dfrac{5}{3}\right)\left(-\dfrac{8}{3}\right)\left(\dfrac{1}{2}\right)^3}{3!}=-\dfrac{5}{81}$	M1A1 [2]
		Total 8 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 2 of 13 · Question 2 part a · 6 marks

2024 JUN · Paper 01 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 3 of 13 · Question 2 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

	$C\sqrt{2}+D$ e.g., $\frac{5+\sqrt{18}}{\sqrt{50}-6}\times\frac{\sqrt{50}+6}{\sqrt{50}+6}$ scores this mark. That is all that is required for this mark. Ignore $>, <, =$ for this mark
dM1	For explicitly multiplying out and simplifying their expression to obtain a value for x in the required form. Allow no more than one error in this simplification. However, it must be simplified as far as $\frac{A+\sqrt{B}}{C}$ or $\frac{P+Q\sqrt{R}}{T}$ where A, B, C, P, Q, R and T are integers Ignore $>, <, =$ for this mark
A1	For obtaining the value of x as given with the correct inequality. For candidates who give an estimated value [8.629….] , please isw if you see the exact value first.

Type MS · Finding unknown constants from binomial coefficients

Answer 3 of 13 · Question 2 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

ALT 1 – Squares both sides [Ignore <, >, = for the first 3 marks]	
M1	Squares both sides without errors. $\left(\sqrt{50}x-\sqrt{18}\right)^2>\left(6x+5\right)^2\Rightarrow 50x^2-60x+18>36x^2+60x+25$
M1	Collects up their like terms and forms a 3TQ $\Rightarrow 14x^2-120x-7>0$
dM1	Solves their 3TQ by either valid method using the formula or completing the square. $x=\frac{60\pm 43\sqrt{2}}{14}$ oe eg., $x=\frac{60\pm\sqrt{3698}}{14}$ Accept $\pm$ for this mark We MUST see a method for the award of this mark. Do not award for roots appearing with no working. Ignore >, <, = for this mark
A1	For obtaining the value of x as given with the correct inequality. This has to be $x>\frac{43\sqrt{2}+60}{14}$ For candidates who give an estimated value [8.629….] , please isw if you see the exact value first.
ALT 2 – Collects up like terms and squares [Ignore <, >, = for the first 3 marks]	
M1	Squares both sides without error. $\left(\sqrt{50}x-6x\right)^2>\left(5+\sqrt{18}\right)^2\Rightarrow 50x^2-12\sqrt{50}x^2+36x^2>25+10\sqrt{18}+18$
M1	Collects up like terms and forms a **2TQ** $86x^2-12\sqrt{50}x^2-\left(43+10\sqrt{18}\right)>0\Rightarrow\left[\left(86-60\sqrt{2}\right)x^2-\left(43+30\sqrt{18}\right)>0\right]$
dM1	Solves their 2TQ by either valid method using the formula or completing the square. $x=\frac{43\sqrt{2}\pm 60}{14}$ oe eg., $x=\frac{\sqrt{3698}\pm 60}{14}$ We MUST see a method for the award of this mark. Do not award for roots appearing with no working.
A1	For obtaining the value of x as given with the correct inequality. This has to be $x>\frac{43\sqrt{2}+60}{14}$ For candidates who give an estimated value [8.629….] , please isw if you see the exact value first.

Type MS · Finding unknown constants from binomial coefficients

Answer 3 of 13 · Question 2 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks
2(a)	$(1+Ax)^n = 1+n(Ax)+\frac{n(n-1)(Ax)^2}{2}+\ldots$	B1
	$\Rightarrow nA=-\frac{1}{3}$ $\quad \frac{n(n-1)A^2}{2}=\frac{5}{36}$	M1
	$\Rightarrow A=-\frac{1}{3n}$	
	$\Rightarrow \frac{n(n-1)\left(-\frac{1}{3n}\right)^2}{2}=\frac{5}{36} \Rightarrow n=-\frac{2}{3} \Rightarrow A=\frac{1}{2}$	M1dM1A1A1
	ALT 1	[6]
	$\Rightarrow nA=-\frac{1}{3}$ $\quad \frac{n(n-1)A^2}{2}=\frac{5}{36}$	[M1
	$\Rightarrow n=-\frac{1}{3A}$	
	$\Rightarrow \frac{-\frac{1}{3A}\left(-\frac{1}{3A}-1\right)(A)^2}{2}=\frac{5}{36} \Rightarrow A=\frac{1}{2} \Rightarrow n=-\frac{2}{3}$	M1dM1A1A1]
	ALT 2	
	$\Rightarrow nA=-\frac{1}{3} \Rightarrow \left[(nA)^2=\frac{1}{9}\right]$	
	$\frac{n(n-1)A^2}{2!}=\frac{5}{36} \Rightarrow \left[\frac{(nA)^2-A(An)}{2!}=\frac{5}{36}\right]$	[M1
	$\Rightarrow \frac{\frac{1}{9}-A\left(-\frac{1}{3}\right)}{2}=\frac{5}{36} \Rightarrow A=\frac{1}{2} \Rightarrow n=-\frac{2}{3}$	M1dM1A1A1]
(b)	Coefficient of $x^3 \Rightarrow \frac{\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)\left(-\frac{8}{3}\right)\left(\frac{1}{2}\right)^3}{3!}=-\frac{5}{81}$	M1A1 [2]
		Total 8 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 3 of 13 · Question 2 part b · 2 marks

2024 JUN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 4 of 13 · Question 5 part a · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number		
5 (a)	$1+anx+\frac{n(n-1)}{2}a^2x^2+\frac{n(n-1)(n-2)}{3!}a^3x^3$	M1 A1 (2)
(b)	$an=15$	B1
	$\frac{n(n-1)}{2}a^2=\frac{n(n-1)(n-2)}{6}a^3\Rightarrow\left(3=(n-2)a\right)$	M1A1
	Solving simultaneously leading to $a=\ldots$ or $n=\ldots$	M1
	$a=6$ and $n=2.5$	A1 A1 (6)
(c)	$\frac{'2.5'\times1.5\times0.5\times'6'^3}{6}=67.5$	M1 A1 (2)
		Total 10 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 4 of 13 · Question 5 part a · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 5 of 13 · Question 5 part c · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Question number		
5 (a)	$1+anx+\frac{n(n-1)}{2}a^2x^2+\frac{n(n-1)(n-2)}{3!}a^3x^3$	M1 A1 (2)
(b)	$an=15$	B1
	$\frac{n(n-1)}{2}a^2=\frac{n(n-1)(n-2)}{6}a^3\Rightarrow\left(3=(n-2)a\right)$	M1A1
	Solving simultaneously leading to $a=\ldots$ or $n=\ldots$	M1
	$a=6$ and $n=2.5$	A1 A1 (6)
(c)	$\frac{'2.5'\times1.5\times0.5\times'6'^3}{6}=67.5$	M1 A1 (2)
		Total 10 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 5 of 13 · Question 5 part c · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 6 of 13 · Question 5 part a · 2 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

5 (a)	$1+anx+\frac{n(n-1)}{2}a^2x^2+\frac{n(n-1)(n-2)}{3!}a^3x^3$	M1 A1 [2]
(b)	$an=8 \qquad \frac{n(n-1)}{2}a^2=30$	
	$\frac{n(n-1)}{2}\left(\frac{8}{n}\right)^2=30$ or $\frac{\left(\frac{8}{a}\right)\left(\frac{8}{a}-1\right)}{2}a^2=30$	M1
	$32n-32=30n \rightarrow n=$ or $64-8a=60 \rightarrow a=$	dM1
	$n=16 \qquad a=\frac{1}{2}$	A1 A1 [4]
(c)	$\frac{"16"\times("16"-1)("16"-2)}{3!}\times\left("\frac{1}{2}"\right)^3=70$	M1 A1 [2]
		Total 8 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 6 of 13 · Question 5 part a · 2 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Type MS · Finding unknown constants from binomial coefficients

Answer 6 of 13 · Question 5 part a · 2 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 7 of 13 · Question 5 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

Question number		
5 (a)	$1+anx+\frac{n(n-1)}{2}a^2x^2+\frac{n(n-1)(n-2)}{3!}a^3x^3$	M1 A1 [2]
(b)	$an=8 \qquad \frac{n(n-1)}{2}a^2=30$	
	$\frac{n(n-1)}{2}\left(\frac{8}{n}\right)^2=30$ or $\frac{\left(\frac{8}{a}\right)\left(\frac{8}{a}-1\right)}{2}a^2=30$	M1
	$32n-32=30n \rightarrow n=$ or $64-8a=60 \rightarrow a=$	dM1
	$n=16 \qquad a=\frac{1}{2}$	A1 A1 [4]
(c)	$\frac{"16"\times("16"-1)("16"-2)}{3!}\times\left("\frac{1}{2}"\right)^3=70$	M1 A1 [2]
		Total 8 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 7 of 13 · Question 5 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Type MS · Finding unknown constants from binomial coefficients

Answer 7 of 13 · Question 5 part b · 4 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 8 of 13 · Question 5 part c · 2 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

5 (a)	$1+anx+\frac{n(n-1)}{2}a^2x^2+\frac{n(n-1)(n-2)}{3!}a^3x^3$	M1 A1 [2]
(b)	$an=8 \qquad \frac{n(n-1)}{2}a^2=30$	
	$\frac{n(n-1)}{2}\left(\frac{8}{n}\right)^2=30$ or $\frac{\left(\frac{8}{a}\right)\left(\frac{8}{a}-1\right)}{2}a^2=30$	M1
	$32n-32=30n \rightarrow n=$ or $64-8a=60 \rightarrow a=$	dM1
	$n=16 \qquad a=\frac{1}{2}$	A1 A1 [4]
(c)	$\frac{"16"\times("16"-1)("16"-2)}{3!}\times\left("\frac{1}{2}"\right)^3=70$	M1 A1 [2]
		Total 8 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 8 of 13 · Question 5 part c · 2 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Type MS · Finding unknown constants from binomial coefficients

Answer 8 of 13 · Question 5 part c · 2 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 9 of 13 · Question 6 part b · 4 marks

2024 NOV · Paper 01 · find part b in this mark scheme extract

6(a)	eg $\frac{a}{\sqrt{4\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{\sqrt{4}\sqrt{\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{4^{\frac{1}{2}}\left(1+\frac{bx}{4}\right)^{\frac{1}{2}}}$ or $a\left(4^{-\frac{1}{2}}\right)\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$ or $a\left(4\left(1+\frac{bx}{4}\right)\right)^{-\frac{1}{2}}$ $=\frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$	M1 A1*cso [2]
(b)	$\left(\frac{a}{2}\right)\left[1+\left(-\frac{1}{2}\right)\left(\frac{bx}{4}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{bx}{4}\right)^2+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{3!}\left(\frac{bx}{4}\right)^3\right]$ oe $\left(\left(\frac{a}{2}\right)\left[1+\left(-\frac{bx}{8}\right)+\frac{3}{128}b^2x^2+\left(-\frac{5}{1024}b^3x^3\right)\right]\right)$ Not necessary - but must be checked if seen All intermediate steps must be checked for errors $\frac{a}{2}-\frac{ab}{16}x+\frac{3ab^2}{256}x^2-\frac{5ab^3}{2048}x^3$ $P=\frac{a}{2}, Q=-\frac{ab}{16}*, R=\frac{3ab^2}{256}, S=-\frac{5ab^3}{2048}*$	M1 M1 A1 A1*cso [4]
(c)	$-\frac{ab}{16}=\frac{128}{5}.\left(-\frac{5ab^3}{2048}\right)$ or $\frac{5}{128}.\left(-\frac{ab}{16}\right)=-\frac{5ab^3}{2048}$ or $-\frac{5ab}{2048}=-\frac{5ab^3}{2048}$ or $-\frac{ab}{16}=-\frac{ab^3}{16}$ $\left(b^2=1\right)b=1*$ $\frac{3a\times1^2}{256}=\frac{9}{256}\Rightarrow a=3*$	M1 A1* B1*cso [3]
(d)	$\frac{\sqrt{6}}{2}=\frac{3}{2}\left(1+\frac{x}{4}\right)^{-\frac{1}{2}}$ or $\frac{\sqrt{6}}{2}=\frac{3}{\sqrt{4+x}}$ or $\frac{3}{\sqrt{6}}=\frac{3}{\sqrt{4+x}}$ or $\sqrt{6}=\sqrt{4+x}$ oe $\Rightarrow x=(2)$ $\frac{3}{2}-\frac{3\times1}{16}\times2+\frac{3\times3\times1^{(2)}}{256}\times2^2-\frac{5\times3\times1^{(3)}}{2048}\times2^3$ 1.207	M1 dM1 A1 [3]
		Total 12 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 10 of 13 · Question 6 part c · 3 marks

2024 NOV · Paper 01 · find part c in this mark scheme extract

6(a)	eg $\frac{a}{\sqrt{4\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{\sqrt{4}\sqrt{\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{4^{\frac{1}{2}}\left(1+\frac{bx}{4}\right)^{\frac{1}{2}}}$ or $a\left(4^{-\frac{1}{2}}\right)\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$ or $a\left(4\left(1+\frac{bx}{4}\right)\right)^{-\frac{1}{2}}$	M1
	$=\frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$	A1*cso [2]
(b)	$\left(\frac{a}{2}\right)\left[1+\left(-\frac{1}{2}\right)\left(\frac{bx}{4}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{bx}{4}\right)^2+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{3!}\left(\frac{bx}{4}\right)^3\right]$ oe	M1 M1 A1
	$\left(\left(\frac{a}{2}\right)\left[1+\left(-\frac{bx}{8}\right)+\frac{3}{128}b^2x^2+\left(-\frac{5}{1024}b^3x^3\right)\right]\right)$ Not necessary - but must be checked if seen All intermediate steps must be checked for errors $\frac{a}{2}-\frac{ab}{16}x+\frac{3ab^2}{256}x^2-\frac{5ab^3}{2048}x^3 \quad P=\frac{a}{2}, Q=-\frac{ab}{16}*, R=\frac{3ab^2}{256}, S=-\frac{5ab^3}{2048}*$	A1*cso [4]
(c)	$-\frac{ab}{16}=\frac{128}{5}.\left(-\frac{5ab^3}{2048}\right)$ or $\frac{5}{128}.\left(-\frac{ab}{16}\right)=-\frac{5ab^3}{2048}$ or $-\frac{5ab}{2048}=-\frac{5ab^3}{2048}$ or $-\frac{ab}{16}=-\frac{ab^3}{16}$	M1
	$\left(b^2=1\right)b=1*$	A1*
	$\frac{3a\times1^2}{256}=\frac{9}{256}\Rightarrow a=3*$	B1*cso [3]
(d)	$\frac{\sqrt{6}}{2}=\frac{3}{2}\left(1+\frac{x}{4}\right)^{-\frac{1}{2}}$ or $\frac{\sqrt{6}}{2}=\frac{3}{\sqrt{4+x}}$ or $\frac{3}{\sqrt{6}}=\frac{3}{\sqrt{4+x}}$ or $\sqrt{6}=\sqrt{4+x}$ oe $\Rightarrow x=(2)$	M1
	$\frac{3}{2}-\frac{3\times1}{16}\times2+\frac{3\times3\times1^{(2)}}{256}\times2^2-\frac{5\times3\times1^{(3)}}{2048}\times2^3$	dM1
	1.207	A1 [3]
		Total 12 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 11 of 13 · Question 3 part b · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

3(a)	$\left(\ldots\right)\left(\ldots\right)\left(\ldots\right)\left(\ldots\right)\left(\ldots\right)$ 2! 3! $+\frac{(-5)(-6)(-7)(-8)(px)^4}{4!}+\ldots$ $=1-5px+15p^2x^2-35p^3x^3+70p^4x^4+\ldots$	M1 A1A1 (3)
(b)	$70p^4+2\times35p^3=0$ $p=-1$	M1 A1 (2) [5]

(a)	
M1	Attempt the binomial expansion up to and including the term in x^4. Must start with 1 and (px) must appear in at least one term. Ignore terms beyond x^4. 2! or 2, 3! or 6, 4! or 24 accepted.
A1	Any 2 correct algebraic terms, simplified (1 is not algebraic) Numbers must be simplified but $(px)^n$, $n = 2,3,4$ allowed
A1	Fully correct simplified expansion as shown but allow terms such as +(- 5px) etc
(b)	
M1	Use their coefficients and the **given** equation to form an equation in p (If powers of x included give M0)
A1	Correct value of p $p=-1$ only Must have come from correct working

Type MS · Finding unknown constants from binomial coefficients

Answer 11 of 13 · Question 3 part b · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	16	

Type MS · Finding unknown constants from binomial coefficients

Answer 12 of 13 · Question 9 part d · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

9(a)	$(1+2x)^{-\frac{1}{3}} = 1+\left(-\frac{1}{3}\right)(2x)+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)(2x)^2}{2!}+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)\left(-\frac{7}{3}\right)(2x)^3}{3!}+\ldots$ $=1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}$	M1 A1A1 [3]
(b)	$-\frac{1}{2} < x \leqslant \frac{1}{2}$ accept $-\frac{1}{2} < x < \frac{1}{2}$	B1 [1]
(c)	$f(x) = (2+kx^2)\left(1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}\right)$ $=2-\frac{4x}{3}+\frac{16}{9}x^2+kx^2-\frac{224}{81}x^3-\frac{2k}{3}x^3$	M1 M1A1 [3]
(d)	$-\frac{224}{81}-\frac{2k}{3}=-\frac{8}{3} \Rightarrow k=-\frac{4}{27}$	M1A1 [2]
(e)	$\int_{0.1}^{0.2}\left(2-\frac{4x}{3}+\frac{44x^2}{27}-\frac{8}{3}x^3\right)dx=\left[2x-\frac{4x^2}{2\times3}+\frac{44x^3}{3\times27}-\frac{8x^4}{4\times3}\right]_{0.1}^{0.2}$ $=\left(2\times0.2-\frac{2\times0.2^2}{3}+\frac{44\times0.2^3}{81}-\frac{2}{3}\times0.2^4\right)-\left(2\times0.1-\frac{2\times0.1^2}{3}+\frac{44\times0.1^3}{81}-\frac{2}{3}\times0.1^4\right)$ $=0.1828$ NB: Calculator value is 0.18301744	B1FT M1A1ft M1 A1 [5]
		Total 14 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 12 of 13 · Question 9 part d · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt at binomial expansion. • The first term is 1 • The denominators are correct. • The powers of x are correct.
	A1	For at least one algebraic term correct and simplified.
	A1	For a fully correct and simplified expansion.
(b)	B1	For the correct range. Accept $\lvert x\rvert < \frac{1}{2}$ Do not accept $-\frac{1}{2} \leq x \leq \frac{1}{2}$
(c)	M1	For showing the intent to multiply their expansion by $\left(2+kx^2\right)$
	M1	For expanding the two brackets up to terms in x^3. Ignore terms in higher powers.
	A1	For the correct expansion. This does not have to be simplified. Must be in ascending powers, but accept $\frac{16}{9}x^2$ and kx^2 in either order and accept $-\frac{224}{81}x^3$ and $-\frac{2k}{3}x^3$ in either order. $f(x) = 2 - \frac{4x}{3} + x^2\left(\frac{16}{9}+k\right) + x^3\left(-\frac{224}{81}-\frac{2k}{3}\right)$ ISW errors in collection of coefficients once correct expansion in ascending powers seen.
(d)	M1	For setting their coefficient of x^3 equal to $-\frac{8}{3}$
	A1	For the correct value of k
(e)	B1FT	For substituting in their k into coefficient for x^2 in their expansion from (c) and having the coefficient of x^3 as $-\frac{8}{3}$. If the coefficient of x^2 is not correct then the substitution must be seen to award this mark. Note: this is an M mark in epen.
	M1	For an attempt to integrate their expansion which must have 4 terms. At least two powers of x to increase by 1 (including $c \rightarrow cx$). No power of x to reduce.
	A1ft	For a correctly integrated expression follow through their 4 term polynomial. Simplified integrated expression is $2x - \frac{2x^2}{3} + \frac{44x^3}{81} - \frac{2}{3}x^4$
	M1	For substituting in 0.2 and 0.1 the correct way around in their integrated expression. Expression should not have been differentiated and needs to have changed. Not dependent on first M. Must show the substitution if their integral or their limits are incorrect. Correct awrt 0.1828 following correct integral with correct limits implies this mark.
	A1	For the correct value of awrt 0.1828 NB: The calculator value is 0.18301744

Type MS · Finding unknown constants from binomial coefficients

Answer 12 of 13 · Question 9 part d · 2 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding unknown constants from binomial coefficients

Answer 13 of 13 · Question 7 part c · 2 marks

2024 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

7(a)	$\left(1+2x^2\right)^{-\frac{3}{4}} = 1+\left(-\frac{3}{4}\right)\left(2x^2\right)+\frac{\left(-\frac{3}{4}\right)\left(-\frac{7}{4}\right)\left(2x^2\right)^2}{2!}+\frac{\left(-\frac{3}{4}\right)\left(-\frac{7}{4}\right)\left(-\frac{11}{4}\right)\left(2x^2\right)^3}{3!}$ $= 1-\frac{3}{2}x^2+\frac{21}{8}x^4-\frac{77}{16}x^6$	M1 A1A1 [3]
(b)	$f(x) = (2+kx)\left(1-\frac{3}{2}x^2+\frac{21}{8}x^4-\frac{77}{16}x^6\right)$ $= 2+kx-3x^2-\frac{3k}{2}x^3+\frac{21}{4}x^4+\frac{21k}{8}x^5$	 M1A1 [2]
(c)	$14\times(-3) = \frac{21k}{8} \Rightarrow k = \frac{14\times(-3)\times 8}{21} = -16$	M1A1 [2]
		Total 7 marks

Type MS · Finding unknown constants from binomial coefficients

Answer 13 of 13 · Question 7 part c · 2 marks

2024 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract